EXPLAINABLE REVIEWED-ITEM RETRIEVAL, WITH A THAI TASTE CASE STUDY

Progress Report

(Proposal title: *Explainable Reviewed-Item Retrieval: Searching Restaurants by Taste Preference*)

by

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CHAPTER 1

RESPONSE TO PROPOSAL FEEDBACK

1.1 Feedback Summary Page

The table below lists every comment we received on the proposal and what we did about it. We refer to the four concerns as F1 to F4. Sections 1.2 to 1.6 explain each answer in more detail.

Table 1.1: Summary of the proposal feedback and our response.

Comment	**What we did**	**Status**
Strengths: problem formulation, RIRD labels, EDA and baselines, honest split between English and Thai	We kept the reviewed-item retrieval formulation and RIRD as our only restaurant-level benchmark, and we still only claim what we can measure.	Kept
F1. The application claims do not match the evaluation. RIRD is not about Thai taste, and Wongnai cannot support restaurant-level evaluation.	We changed the title and narrowed our claims. We are building a small Thai taste test set: 24 Thai queries over real Wongnai review sentences, to be labelled by native speakers. Thai restaurant ranking is now listed as future work.	Queries, retrieval and labelling guide are ready; labelling comes next
F2. BM25 and dense retrieval use different units, and fusion may be biased by the number of reviews.	Both models now score the same unit (a single review) and use the same aggregation. We also capped every restaurant at 400 reviews to test the volume effect. The bias shows up in the scores, but removing it barely changes the results.	Done at review level; sentence level next

Comment	What we did	Status
F3. There is no evidence that the shown sentences are good explanations.	We split this into two questions: does the ranking really depend on the shown sentences (a deletion test), and do people agree that they support the query (a human rating)? We already found clear failure cases while inspecting the output.	Tests designed; to be run next
F4. Check more literature.	We read and added the three suggested papers, plus work on aggregation bias and on how to evaluate explanations.	Done
Final question: do the retrieved sentences really support the rankings, and does the approach work on real Thai reviews with independent taste labels?	We adopted it as our main research question. The first part will be answered by the two explanation tests, the second by the Thai test set, at sentence level.	Plan fixed
Earlier advice to prefer basics over complexity	We dropped the flavour-feature layer, the extra datasets, rating-weighted fusion and the three-encoder comparison (Section 1.7).	Done

1.2 F1: Our Claims Did Not Match What We Measured

We agree with this comment. The proposal talked about a Thai food recommender, but every number in it came from RIRD, which is a set of English Yelp reviews about restaurants in Toronto. The RIRD queries are general preferences such as "I am on a budget", so none of our results said anything about whether the system understands Thai taste words like เปรี้ยว (sour), เผ็ด (spicy) or ไม่หวานมาก (not too sweet). Our Thai dataset, Wongnai, also has no restaurant names or ids, so it cannot be used to rank restaurants at all.

We changed two things. First, we narrowed the claim. The project is now called *Explainable reviewed-item retrieval, with a Thai taste case study.* Restaurant ranking and explanations are only claimed for English, where RIRD gives us real relevance labels. For Thai we only claim the first step, finding the right review sentences for a taste query.

Second, we started building a way to actually measure the Thai part. We wrote 24 Thai taste queries (11 single tastes, 8 combinations such as sour and spicy, and 5 negations such as "not too sweet"). A native Thai speaker in our group reviewed and fixed the wording before looking at any retrieval results, so the queries could not be tuned to favour one method. Both retrieval methods already run on real Wongnai sentences (Section 5.5), and the next step is to label their results by hand.

We also considered translating RIRD into Thai, but decided against it. A translated review does not read like a real Thai review. Thai diners write แซ่บ, เผ็ดดดด or mix in English words, and with translation we would end up testing the translation system rather than our method.

1.3 F2: Different Units and Review Volume

This comment has two parts. The first is about units. In the proposal, BM25 scored one long document per restaurant (all reviews joined together), while the dense model scored each review and then averaged them. So when dense came out ahead, we could not tell whether the model was better or whether it just had a better setup. We fixed this by making both models score the same thing, a single review, and then combining the review scores into a restaurant score in exactly the same way.

The second part is about volume. In RIRD, restaurants have between 406 and 2,281 reviews. If the restaurant score is the best review score (max), or the average of the best few (top-k mean), then a restaurant with many reviews simply has more chances to contain one very high-scoring review, even if it is not more relevant. A plain average does not have this problem. People working on expert search ran into the same issue, where authors with many documents were ranked too high (Macdonald & Ounis, 2008).

To test this, we capped every restaurant at the same 400 randomly chosen reviews and recomputed all the rankings. The result is in Section 5.3. In short, the bias is real: within a query, max and top-k scores go up with the number of reviews (correlation up to 0.36), and the effect disappears after capping. However, ranking quality changes by at most 0.011 nDCG@10 and our conclusions stay the same. Part of the effect is not really bias either, because restaurants with more reviews are also relevant to more of the RIRD queries.

1.4 F3: Are the Explanations Any Good?

We had not tested this, and the comment made us realise that "good explanation" means two different things (Jacovi & Goldberg, 2020). One is faithfulness: is the shown sentence the real reason the restaurant was ranked high? The other is support: would a person reading the sentence agree that the restaurant fits the query? A sentence can pass one test and fail the other.

Looking at our current output, we found a clear example. For the query "I am on a budget", the top-ranked restaurant is explained by the review *"It's not bad but kinda pricey."* This review really is why the model ranked the restaurant first, because it talks about price, so the explanation is faithful. But it says the opposite of what the user wants, and the RIRD annotators marked this restaurant as not relevant. Section 5.4 shows more examples like this.

We also noticed that the aggregation method matters for explanations. With max or top-k mean, the shown reviews are exactly the ones that produced the score. With a plain average over hundreds of reviews, showing the top three would be misleading, because every review counts equally. This gives us another reason to prefer top-k mean.

We will run two tests (Section 4.2). The deletion test removes the shown sentences and checks how far the restaurant drops, compared with removing random sentences; this follows the comprehensiveness idea from ERASER (DeYoung et al., 2020). In the human rating, two of us label each shown sentence as not supporting, partly supporting or clearly supporting the query, and we report how much we agree. Our rating criteria follow the user-centred view of Zhang et al. (2025). We only select existing review sentences; we do not generate new text.

1.5 F4: More Literature

We added all three suggested papers to Chapter 3. The closest one to our work is Korikov et al. (2024). They show that top-K fusion struggles when a query asks for several things that appear in different reviews, and they solve it by using an LLM to split the query into parts. We cite this but do not reimplement it. Our question is whether fusion is fair, not how to split queries, and an LLM step would add cost and make the results harder to reproduce, especially for Thai. Their finding does suggest that our Thai combination queries, such as อะไรแซ่บๆ เปรี้ยวเผ็ดเค็ม (something zesty, sour, spicy and salty), will be the hardest, so we will report them separately. Zhang et

al. (2025) helped us decide what to look for when rating explanations, and the survey by Hasan et al. (2025) helped us place the project: most review-based recommenders use reviews to predict ratings, while we use them to answer a free-text query.

1.6 The Final Question

The professor summarised the feedback with this question:

> *Do retrieved review sentences genuinely support the system's restaurant rankings, and can the same approach work on real Thai restaurant reviews with independently annotated taste preferences?*

We have made it the main research question of the project. The first half will be answered on RIRD with the deletion test and the human rating. The second half will be answered with the Thai test set, but only at sentence level, since no Thai dataset links reviews to restaurants. At this point neither half has a measured answer yet. What we have is a working pipeline in both languages and a clear plan for getting the labels.

1.7 What We Decided Not to Do

Table 1.2 lists the parts of the proposal we removed and one suggestion we chose not to follow.

Table 1.2: Items removed from the scope, and why.

Item	**Reason**
LLM query decomposition (Korikov et al., 2024)	It answers a different question, adds cost, and is hard to reproduce or apply to Thai. We cite it instead.
Rating-weighted fusion	Star ratings measure overall satisfaction, not relevance to the query, and would add another popularity effect, which goes against F2.
Comparing three encoders	Using one multilingual encoder for both English and Thai keeps the two parts comparable, and frees time for F2 and F3.
Flavour features, Thailand Foods, M-ABSA	These risked validating our own features against themselves, and M-ABSA has no taste-intensity labels. Human labels on Thai sentences replace them.
Fine-tuning	Out of scope; all models are used as they are.

CHAPTER 2

PROJECT OVERVIEW

2.1 Problem

A user types what they feel like eating, for example "I am on a budget" or **อยากกิน อะไรเปรี้ยวๆ** (I want something sour). The system should rank restaurants and show a few review sentences that explain each choice. Every restaurant has many reviews. A retrieval model gives each review (or sentence) a score for the query, and the restaurant score is built from those scores:

$$\text{Score}(q, r) = f\big(\text{scores of the reviews of } r\big), \qquad f \in \{\text{max, mean, mean of the top } k\}. \tag{2.1}$$

The highest-scoring sentences of each restaurant are shown to the user as the explanation.

2.2 Research Questions

RQ1. When BM25 and dense retrieval are compared fairly, on the same unit and with the same aggregation, which one ranks restaurants better, and does the answer hold once review volume is controlled?

RQ2. Do the shown sentences really drive the ranking, and do people agree that they support the query?

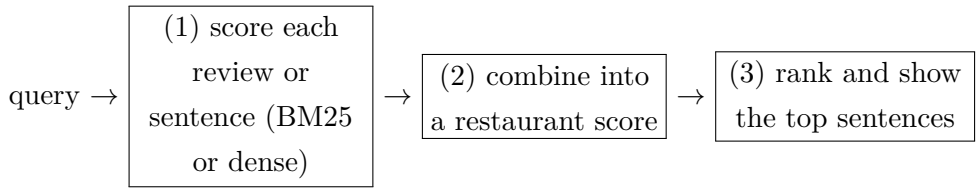
RQ3. Does sentence retrieval work on real Thai restaurant reviews, judged by native speakers?

2.3 Data

Table 2.1: The two datasets used after the revision.

Dataset	Language	Contents	Used for
RIRD (Abdollah Pour et al., 2023)	English	29,168 reviews of 50 restaurants (406 to 2,281 each); 100 queries judged against all 50 restaurants by 5 annotators	RQ1 and RQ2
Wongnai (Highground-bkk, n.d.)	Thai	40,000 reviews with text and a 0–4 star rating; no restaurant ids	RQ3, with our own queries and labels

2.4 Pipeline



RQ1 tests steps 1 and 2, RQ2 tests step 3, and RQ3 tests step 1 in Thai.

Figure 2.1: The three steps of the system and which research question tests each step.

For the dense model we use `paraphrase-multilingual-mpnet-base-v2` (Reimers & Gurevych, 2020) in both languages. For BM25 (Robertson & Zaragoza, 2009) we split English text into lower-case words and Thai text into words with PyThaiNLP (Phatthiyaphaibun et al., 2023), since Thai is written without spaces.

CHAPTER 3

RELATED WORK (UPDATED)

3.1 Reviewed-Item Retrieval

Abdollah Pour et al. (2023) introduced the task we work on, where items are described by their reviews, and released the RIRD dataset. They compare averaging the reviews first with scoring each review and combining the top K scores afterwards, and the second approach works better. Their best model, which is fine-tuned, reaches a MAP of 0.626. They also show that a dense model without fine-tuning already beats BM25, so our own finding on this point repeats theirs with a different, multilingual encoder. Korikov et al. (2024) build on this and show that top-K fusion breaks down when the different parts of a query are spread over different reviews. Neither paper checks for a review-volume effect or tries sentences instead of whole reviews, and that is where our study adds something.

3.2 Combining Scores and the Volume Problem

Turning many document scores into one score for an entity is an older problem from expert search. Balog et al. (2006) compare building one big profile per person, which is similar to the concatenated document in our proposal, with scoring each document and combining the scores, which is what we do now; the second approach was better. Macdonald & Ounis (2008) describe these combinations as voting and point out that people with many documents get an unfair advantage unless the scores are normalised. Our capping experiment is the reviewed-item version of this question. Finally, Chen et al. (2024) find that dense retrievers often work better on smaller pieces of text, which is why we also plan a sentence-level run.

3.3 Evaluating Explanations

Using existing review sentences as explanations is a common approach (Li et al., 2021). Jacovi & Goldberg (2020) argue that faithfulness (does the explanation reflect what the model did) and plausibility (does it convince a person) should be measured

separately. ERASER (DeYoung et al., 2020) measures faithfulness by removing the explanation and seeing how much the model's output changes, and our deletion test borrows this idea. Zhang et al. (2025) look at explanations from the user's side and ask, among other things, whether an explanation is factually backed by the item; this is close to what our human rating asks. The survey by Hasan et al. (2025) names explainability and evaluation as open problems for review-based recommenders.

3.4 Thai Retrieval

Most Thai retrieval benchmarks cover Wikipedia, question answering or legal text. On the Thai part of MIRACL (Zhang et al., 2023), BM25 (nDCG@10 0.484) actually beats an older dense model (0.358), so we do not assume that dense retrieval will win in Thai. Negation is also known to be difficult for neural retrievers (Weller et al., 2024), so we report our Thai negation queries separately. We did not find any existing Thai test set for taste-based search over restaurant reviews.

CHAPTER 4

UPDATED METHODOLOGY

4.1 Fair Comparison on RIRD

We build one BM25 index over all 29,168 reviews and embed every review with the dense model (up to 256 tokens). Both models then give a score to every review for every query, and the same code turns these review scores into restaurant scores using max, mean, or the mean of the top 3, 5 or 10 reviews.

For evaluation we use nDCG@10 as the main measure, with the number of annotators who marked a restaurant relevant (0 to 5) as the graded label (Järvelin & Kekäläinen, 2002). We also report MAP, MRR, precision and recall. Only about 27% of query–restaurant pairs are relevant, so we only use measures that look at the ranking. With 100 queries, small differences can easily be noise, so every score comes with a 95% bootstrap confidence interval, and comparisons between two models use a paired randomization test with Holm correction for multiple comparisons (Smucker et al., 2007).

We also include three simple baselines that ignore the query: a random order, ranking by number of reviews, and ranking by popularity, meaning how many *other* queries a restaurant is relevant for. If a model cannot beat popularity, it is mostly finding restaurants that are good for everything rather than matching the query.

For the volume test we take 400 random reviews from every restaurant (the smallest has 406), recompute the restaurant scores from the same review scores, and compare the results and the correlation between restaurant score and review count. For every query, model and aggregation we also save the full ranking and the top 10 reviews behind each restaurant, so that the explanation tests use exactly what the system produced.

4.2 Testing the Explanations

Deletion test. For each query we take the top three restaurants, remove the 1, 3, 5 or 10 sentences shown as the explanation, recompute the restaurant's score and see where it now ranks. We compare this with removing the same number of random

sentences, repeated 20 times. If the explanation matters, removing it should push the restaurant down much more than removing random sentences. With top-k mean, removing the top sentences always lowers the score, so what we really look at is how far the restaurant falls. If it hardly moves, other sentences are almost as good, which means the explanation is real but not unique.

Human rating. We take 30 queries, the top three restaurants for each and their top three sentences, which gives 270 items. Two of us label each item without seeing which model produced it: 0 if the sentence is unrelated or contradicts the query, 1 if it partly supports it, and 2 if it clearly shows why the restaurant fits. We report Cohen's kappa (Cohen, 1960), the share of sentences rated as supporting, and the most common types of failure.

4.3 Thai Taste Test Set

The 24 Thai queries cover single tastes (sour, spicy, sweet, rich, oily, cheap, salty, herbal and mild), combinations such as sour and spicy tom yum, and negations such as desserts that are not too sweet. They were fixed on 17 October 2026, before any results were looked at.

Reviews are split into sentences with PyThaiNLP's `crfcut`, and fragments shorter than 15 characters are dropped. For the pilot we used 2,000 reviews. For the actual test set we will use 10,000 reviews (about 95,000 sentences), because rare tastes and negations need enough truly relevant sentences in the collection. Otherwise both methods would look bad simply because there is little to find. We fixed this size before labelling, because the labels only make sense for one fixed collection.

For each query we put the top 10 sentences from BM25 and the top 10 from the dense model into one shuffled list, without saying which method found them. A native speaker then labels each sentence 0, 1 or 2 using written rules for every query. For example, for "not too sweet" a sentence saying หวานเกินไป (too sweet) gets 0 and ไม่หวานมาก (not very sweet) gets 2, and for a combination query a sentence with both tastes gets 2 and one taste gets 1. A second Thai speaker labels 60 of the same sentences so we can measure agreement. We will then compute nDCG@10 and precision@10 for both methods, with confidence intervals and separate results for single, combination and negation queries.

CHAPTER 5

PROGRESS AND PRELIMINARY RESULTS

5.1 Work Completed Since the Proposal

Table 5.1: Work completed since the proposal (17 September 2026).

Work	Result	Addresses
Fair comparison	BM25 and dense on the same unit and aggregations, with confidence intervals and significance tests	F2, RQ1
Volume test	Every restaurant capped at 400 reviews	F2, RQ1
Saved evidence	Rankings and top reviews stored for every run	F3, RQ2
Thai queries	24 queries, fixed by a native speaker before any results	F1, RQ3
Thai retrieval	Sentence splitting, BM25 and dense retrieval, first comparison	F1, RQ3
Thai test set design	10,000-review collection, blind pooling and scoring plan	F1, RQ3
Labelling guide	Rules for Thai relevance and for English explanation support	F1, F3
Literature	Related work updated	F4

5.2 Fair Comparison at Review Level

Table 5.2 and Figure 5.1 show all models when both score single reviews.

Table 5.2: Results on RIRD with reviews as the unit (100 queries). nDCG@10 with 95% confidence interval.

Model	**nDCG@10 [95% CI]**	**MAP**	**MRR**	**P@10**	**R@10**
Dense, top-10 mean	**0.668** [0.625, 0.711]	**0.608**	0.760	**0.473**	**0.520**
Dense, top-5 mean	0.668 [0.625, 0.709]	0.606	0.747	0.472	0.518
Dense, top-3 mean	0.665 [0.623, 0.705]	0.601	0.744	0.470	0.517
Dense, mean	0.642 [0.598, 0.687]	0.584	0.750	0.448	0.473
Dense, max	0.625 [0.585, 0.666]	0.564	0.715	0.438	0.480
BM25, top-10 mean	0.634 [0.590, 0.676]	0.576	0.762	0.427	0.476
BM25, top-5 mean	0.621 [0.579, 0.662]	0.566	0.760	0.422	0.473
BM25, top-3 mean	0.611 [0.568, 0.654]	0.555	0.749	0.420	0.461
BM25, max	0.575 [0.532, 0.618]	0.519	0.691	0.404	0.433
BM25, mean	0.549 [0.503, 0.594]	0.507	0.636	0.377	0.423
Dense, pooled vector (proposal)	0.645 [0.602, 0.687]	0.585	0.759	0.449	0.480
BM25, concatenated doc (proposal)	0.621 [0.578, 0.663]	0.561	**0.769**	0.407	0.450
Popularity	0.484 [0.426, 0.540]	0.426	0.436	0.410	0.341
Review count	0.374 [0.324, 0.424]	0.349	0.505	0.281	0.202
Random	0.348 [0.311, 0.387]	0.322	0.428	0.270	0.203

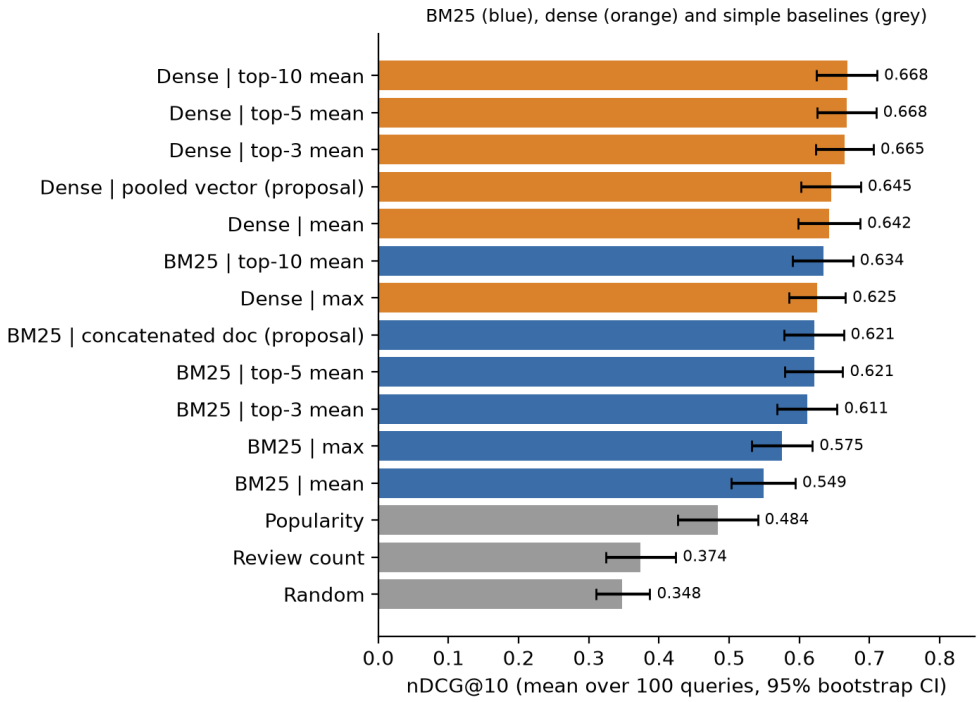


Figure 5.1: nDCG@10 with 95% confidence intervals for all review-level models and the three simple baselines.

Table 5.3: Dense minus BM25 (nDCG@10) for each aggregation, with Holm-corrected p-values from a paired randomization test.

Aggregation	**Difference [95% CI]**	p **(Holm)**	**Significant**
max	+0.050 [0.017, 0.083]	0.009	yes
mean	+0.094 [0.057, 0.129]	<0.001	yes
top-3 mean	+0.053 [0.023, 0.083]	0.003	yes
top-5 mean	+0.047 [0.016, 0.077]	0.009	yes
top-10 mean	+0.035 [0.005, 0.063]	0.020	yes

Dense retrieval beats BM25 with every aggregation, and all five differences are significant (Table 5.3). The gap is largest with the plain mean (+0.094) and smallest with the top-10 mean (+0.035). We think this is because most reviews share no words with the query, so BM25 gives them a score of zero and averaging over all of them washes out the few good matches. Cosine similarity is never exactly zero, so the dense model suffers less.

For both models, the mean of the top few reviews works best and taking only the single best review works worst for dense. One review on its own is a noisy signal, while the average of the best five or ten still focuses on the relevant reviews but is less affected by one odd review.

All models except BM25 with the plain mean beat the popularity baseline (for BM25 mean the difference is +0.065 with $p = 0.056$). Popularity alone already reaches 0.484, which tells us that a large part of the score on RIRD comes from knowing which restaurants suit many queries.

Not everything improved. The fair setup did not give significantly better numbers than the proposal baselines: dense top-10 mean is 0.023 higher than the proposal's pooled vector, but this is not significant ($p = 0.64$), and BM25 with the plain mean is significantly worse than the proposal's concatenated document (-0.072). The aim of the change was a fair comparison rather than a higher score, and the main conclusion from the proposal, that dense beats BM25, still holds.

For context, our best MAP of 0.608 is close to the 0.626 of the fine-tuned model in Abdollah Pour et al. (2023). The models and training are different, so this is not a direct comparison, but it suggests that an off-the-shelf multilingual model is a reasonable starting point for both languages.

5.3 Review-Volume Test

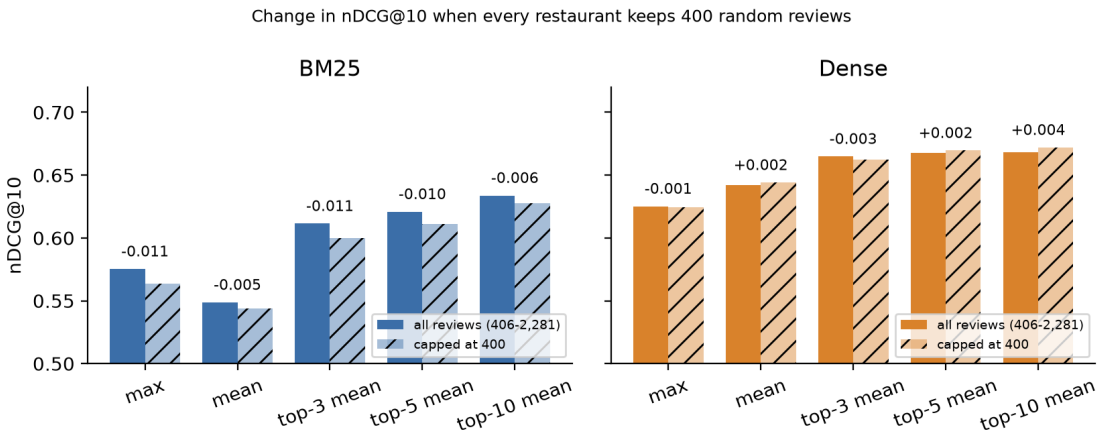


Figure 5.2: nDCG@10 using all reviews versus 400 random reviews per restaurant. The numbers show the change.

Table 5.4: Correlation between restaurant score and number of reviews, averaged over the 100 queries (Pearson / Spearman). "Capped" uses the scores after capping and the original review counts.

	BM25		**Dense**	
Aggregation	**all reviews**	**capped**	**all reviews**	**capped**
max	0.21 / 0.18	0.01 / 0.00	0.22 / 0.11	0.03 / −0.04
mean	−0.05 / −0.06	−0.07 / −0.07	0.04 / −0.06	0.03 / −0.06
top-3 mean	0.28 / 0.23	0.00 / −0.01	0.26 / 0.14	0.04 / −0.04
top-5 mean	0.31 / 0.26	0.00 / −0.01	0.28 / 0.15	0.04 / −0.05
top-10 mean	0.36 / 0.30	−0.01 / −0.02	0.30 / 0.16	0.04 / −0.05

The effect we expected is there. Within a query, restaurants with more reviews get higher max and top-k scores (correlations between 0.21 and 0.36), while the plain mean shows no such effect. After capping, the correlation drops to about zero.

What surprised us is how little this matters for the final rankings. The dense model changes by at most 0.004 nDCG@10 and BM25 loses between 0.005 and 0.011. Dense is still the best with every aggregation. One explanation is that the effect is partly real: restaurants with more reviews are relevant to more RIRD queries (Spearman correlation 0.30 over the 50 restaurants), so preferring them is not entirely wrong. BM25 may lose a little because it depends on exact word matches, and fewer reviews means fewer chances to find them.

We also learned something about how to measure this. When we first pooled all queries into one correlation, we got only 0.05 to 0.12, because the score ranges differ a lot between queries. Averaging the correlation per query, as in Table 5.4, shows the effect more clearly. We used only one random sample of 400 reviews, and RIRD only contains restaurants with at least 400 reviews, so restaurants with very few reviews, where the effect could be stronger, are not covered.

5.4 A First Look at the Explanations

We have not run the two explanation tests yet, but we looked at what the best model currently shows to the user. Table 5.5 gives two examples.

Table 5.5: Top supporting review for the two highest-ranked restaurants (dense, top-10 mean). "Relevant" is the RIRD label.

Query	Rank: restaurant	Relevant	Top review (excerpt)
I am on a budget	1: Fishman Lobster Clubhouse	no	"It's not bad but kinda pricey."
	2: Dragon Boat Fusion	yes	"Good dim sum, pretty standard prices/service."
Non-spicy Chinese meals	1: Ding Tai Fung	yes	A 94-word review about Northern Chinese food (…) "they use a lot of oil" Second review: "…Wonton was spicy … red chili oil"
	2: Pai Northern Thai Kitchen	no	"Perfect Thai Curry in nice easy going atmosphere …"

These examples show the problem the professor pointed out. The dense model finds reviews about the right topic, such as price or spiciness, but it does not notice whether the review agrees with the query. A review saying a place is pricey puts that restaurant first for "I am on a budget", and a review about spicy wontons is shown for "non-spicy". The explanation is faithful to the model but does not support the query.

The examples also show that a whole review is too long to work as an explanation. A 94-word review talks about many things, and the part that matters for the query is hard to find. This is one more reason to repeat the experiments with sentences before the human rating. Based on what we have seen, we expect a fair number of shown sentences to be rated 0, especially for price and negation queries.

5.5 Thai Sentence Retrieval

In the 2,000-review pilot, `crfcut` produced 19,294 sentences, about 9.6 per review. A native speaker checked a sample. Short, casual reviews, which are most of Wongnai, were split well. We saw two kinds of errors: sentences cut after abbreviations such as กทม. or ม.ค. (0.7% of sentences), and long blog-style sections merged into one sentence (1.6% are longer than 200 characters). Short taste statements such as เผ็ด

มาก (very spicy) or ไม่หวานมาก (not very sweet) almost always stayed intact, so we think the splitting is good enough for retrieval. We will mention the published accuracy of `crfcut` on review text as a limitation (Lowphansirikul et al., 2020).

Table 5.6: Top three Thai sentences for one query of each type (2,000-review pilot, inspected but not yet labelled).

Query	BM25	Dense
T02 อยากกินอะไรเผ็ดๆ แบบจัดๆ (really spicy)	เนื่องจากอยากกินปูเน้นๆเนื้อๆ เลยจัดเต็ม อยากกินอะไรเบาๆและชิมชานมเย็น ถ้าอยากกินไอติมแบบอร่อย นั่งคุยกันสบายๆ	ถ้าชอบเผ็ดสั่งเพิ่มได้เลย สำหรับคนชอบเผ็ดๆเลย ต้องกินกับซอสที่ออกเปรี้ยวๆ แซลมอนรมควัน
T11 ต้มยำเปรี้ยวๆ เผ็ดๆ (sour and spicy tom yum)	น้ำต้มยำอร่อยๆ เปรี้ยวๆ ดี อร่อยดี เปรี้ยวๆ เผ็ดๆ แต่ไม่เข็ดฟันเด้อ - พิซซ่าญี่ปุ่น เปรี้ยวอย่างเดียวเปรี้ยวเผ็ดเกิน	เผ็ดเปรี้ยวนำ - ลาบหมู น้ำต้มยำเปรี้ยวแซ่บ หอมกลิ่นพริกขี้หนู ยำหมูยอแซ่บซี้ด
T20 ของหวานที่ไม่หวานมาก (desserts not too sweet)	ตอบโจทย์วัยรุ่นที่อยากกินของหวานแต่ไม่ชอบแบบหวานเลี่ยน ไม่หวานมากอย่างที่คิดไว้ ไม่หวานมากกำลังดี	และมีความหวานน้อย ขนมถ้วย : ไม่หวานมาก ไม่หวานมากแล้วก็ไม่ฝาดมาก ปล.

Without labels we can only describe what we see, but the two methods fail in different ways. For T02, BM25 picks up the filler words อยากกินอะไร ("want to eat something") and จัด instead of the taste, while the dense model finds sentences about spicy food. The dense model has its own problem: for T11 it returns sour and spicy sentences about other dishes, such as larb and yam, and its fifth result for T02 describes a mild taste (รสชาติจะออกอ่อนๆ), which is the opposite of the query. Both methods handle the negation query T20 well, which is different from what is reported for English (Weller et al., 2024). One possible reason is that BM25 keeps ไม่ (not) as a word and the phrase ไม่หวานมาก is very common in Thai reviews. The labels will tell us whether this holds.

The two methods also rarely agree. On average their top 10 lists share less than one sentence per query, so pooling them gives about 19 sentences per query to label (465 in the pilot). That means more labelling work, but also a more informative comparison.

5.6 Challenges

The dense model reads at most 256 tokens, and 19.9% of RIRD reviews are longer than that (in the proposal the limit was 128 tokens, which cut 50.6% of reviews). Part of every long review is therefore never seen, which is another argument for working with sentences.

Thai relevance can only be judged by native speakers, and we need a second independent Thai annotator to measure agreement. Finally, 24 Thai queries is a small test set. With this many queries we can only reliably detect differences of roughly 0.11 to 0.17 nDCG@10, so if we find no significant difference we will report it as "no evidence of a difference", not as the two methods being equal.

CHAPTER 6

REMAINING WORK AND PLAN

Table 6.1: Remaining work by week after the progress presentation (week 1 starts on 21 October 2026).

Work	Description	Week
Sentence-level run	Repeat the RIRD comparison with sentences instead of reviews; repeat the volume test with more random samples and a significance test	1
Thai pool	Run both methods on the 10,000-review collection and build the blind labelling sheet	1
Labelling guide	Final check of the Thai labelling rules by a native speaker	1
Deletion test	Remove the shown sentences ($m = 1, 3, 5, 10$) and compare with random removal	2
Thai labelling	Label the pooled sentences (about 450)	2
Human rating	Two annotators rate 270 explanation items; compute kappa	3
Thai agreement	A second native speaker labels 60 sentences; compute kappa	3
Thai results	nDCG@10 and precision@10 for BM25 and dense, by query type	3
Final write-up	Final report, slides and video	4

By the end we expect to answer RQ1 at both review and sentence level. We expect dense to stay ahead, but we do not know yet whether sentences will help. For RQ2 we will report two numbers per model: how far restaurants drop when their explanation is removed, compared with random removal, and how many of the shown sentences people rate as supporting. If that second number turns out low, it is still a useful finding and we will report it as such. For RQ3 we will report Thai nDCG@10 for both methods with confidence intervals and a separate result for negation queries. Given the MIRACL results, we do not expect either method to clearly win.

If we run short of time, we will first reduce the Thai test set to 15 queries and then the human rating to 15 queries. We will not drop the sentence-level run or the deletion test, because without them we cannot answer the final question.

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